Zinc Finger Nuclease (ZFN) Design for BCL11A

What we did, what we obtained and how to report it

Region: BCL11A, chr2:60462161-60462397 (hg38). Sequence source: UCSC Genome Browser. Access date: 24 October 2026. All designs are predicted or candidate designs; nothing was built or tested.

1. Goal

The aim was to add a zinc finger nuclease (ZFN) design to the CRISPR and TALEN designs already made for the same BCL11A exon, and to show why ZFNs are the hardest of the three to design. A ZFN is a pair of proteins. Each is a zinc finger array fused to a FokI nuclease domain. Each finger recognises 3 bp, so a 3-finger array reads a 9 bp half-site. FokI cuts only when the two arrays bind opposite strands separated by a short spacer (5-7 bp was used here).

2. Summary of the outcome

- The usual design web tool could not be used: the page we reached returned **403 Forbidden** (Figure Z1). We could not confirm that ZiFiT itself is online.
- To continue, we copied the 237 bp BCL11A amplicon sequence from UCSC (Figures Z2-Z4) and scanned it by hand.
- Using the GNN rule (three triplets that each start with G) with a 5-7 bp spacer, **only one candidate pair** exists in the 237 bp region (Table 1, Figure Z5).
- This pair lies close to the CRISPR guide (chr2:60462265) and the TALEN pair (chr2:60462267), so the three designs are in the same neighbourhood.
- The finger module names were **not** assigned, because they must come from the Zinc Finger Consortium archive or the original modular-assembly paper.

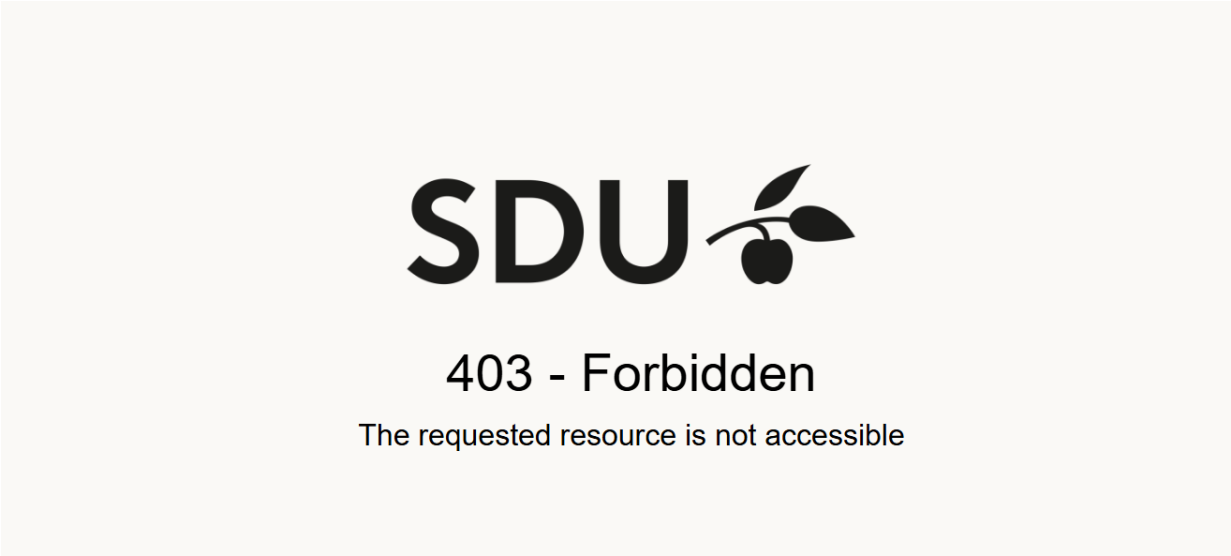
3. What we did, step by step

Step	Action	Result
1	Planned the ZFN design: two 9 bp half-sites on opposite strands, 5-7 bp spacer, same exon as the CRISPR and TALEN designs.	Design rules fixed (3 fingers per side).
2	Tried to open a ZFN design tool link.	403 Forbidden page (Figure Z1). Tool not usable from that link.
3	Looked for a way to get the target DNA instead. Opened UCSC Genome Browser, assembly hg38.	Gene view with the CHOPCHOP target track (Figure Z2).
4	Typed chr2:60462161-60462397 in the position box.	237 bp region displayed (Figure Z3).

Step	Action	Result
5	Requested the DNA text from the UCSC sequence service for the same coordinates.	237 nt sequence returned (Figure Z4).
6	Checked the sequence: it contains the TALEN site and the CRISPR spacer used earlier; the right primer is at its start.	Correct region and orientation confirmed.
7	Scanned both strands for pairs of GNN-GNN-GNN half-sites with a 5, 6 or 7 bp spacer.	One pair found (Table 1).
8	Converted the pair to genomic coordinates and compared them with the CRISPR and TALEN sites.	Pair lies within about 30 bp of both.

4. Figures and explanations

Figure Z1: The design-tool page that was blocked

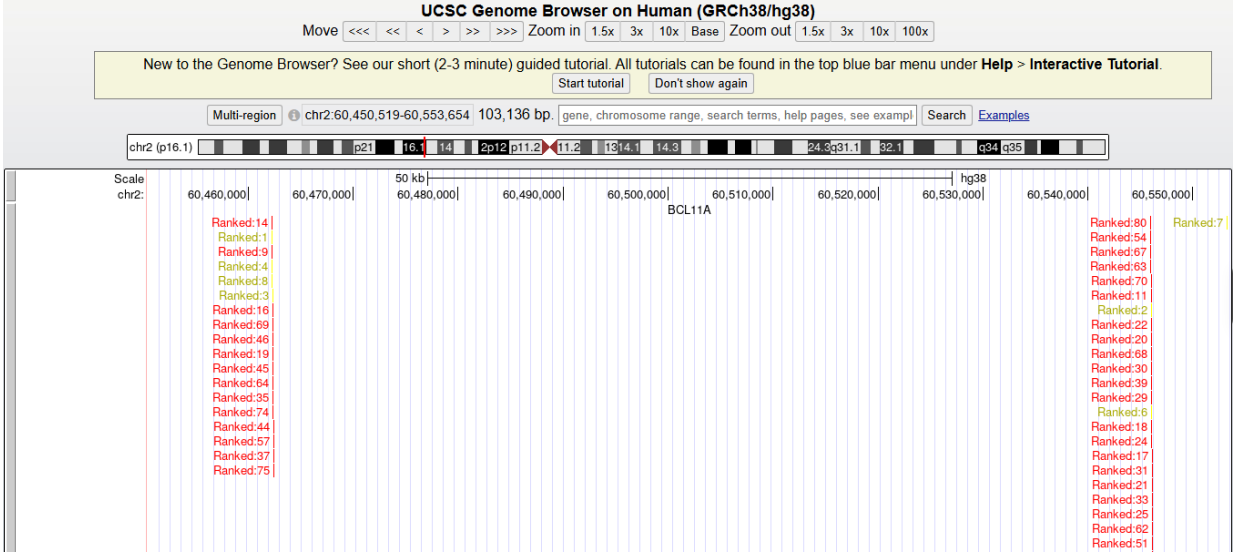


Caption. A university-hosted page returned 403 Forbidden when the design tool link was opened. Seen 24 Oct 2026.

What it shows. The server refused access: 'The requested resource is not accessible'. This is a server-side block, not an input error. We did not identify which tool the link belonged to.

Why it matters. It documents why the standard web tool was not used, which the paper should state plainly. It also explains why a manual design was done.

Figure Z2: UCSC Genome Browser, whole BCL11A region (hg38)

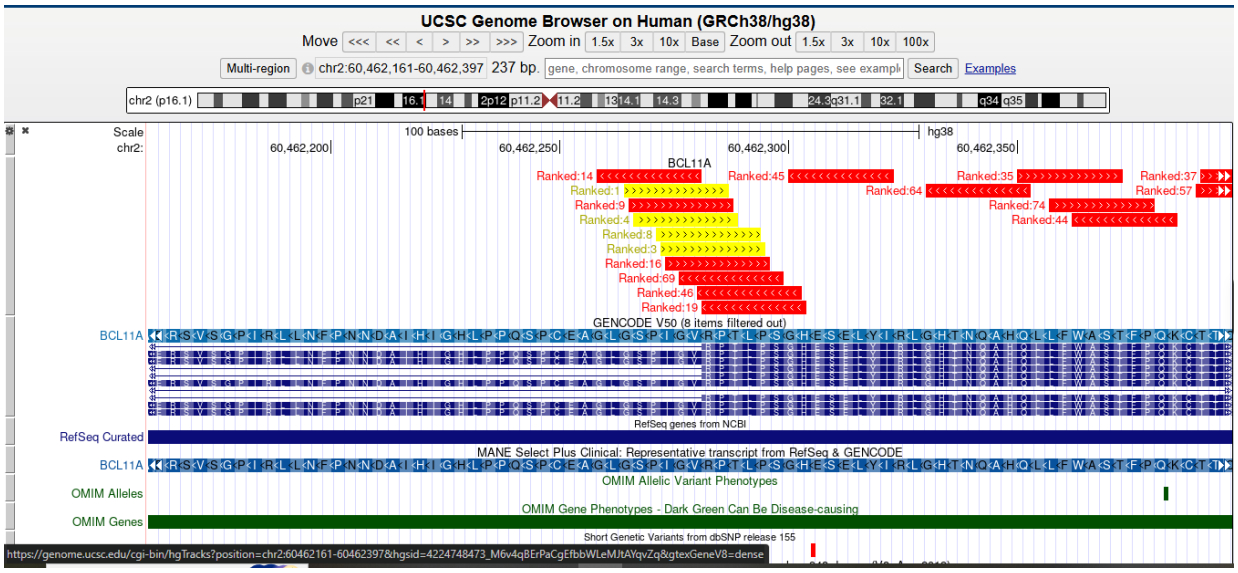


Caption. UCSC Genome Browser on Human (GRCh38/hg38), position chr2:60,450,519-60,553,654 (103,136 bp), showing the ranked CHOPCHOP target track.

What it shows. The assembly is GRCh38/hg38, matching the CHOPCHOP design. The ranked targets form two clusters, one near chr2:60,460,000 and one near chr2:60,545,000.

Why it matters. It confirms the correct genome build before coordinates were used. Using the wrong build would give a different sequence.

Figure Z3: The 237 bp amplicon region

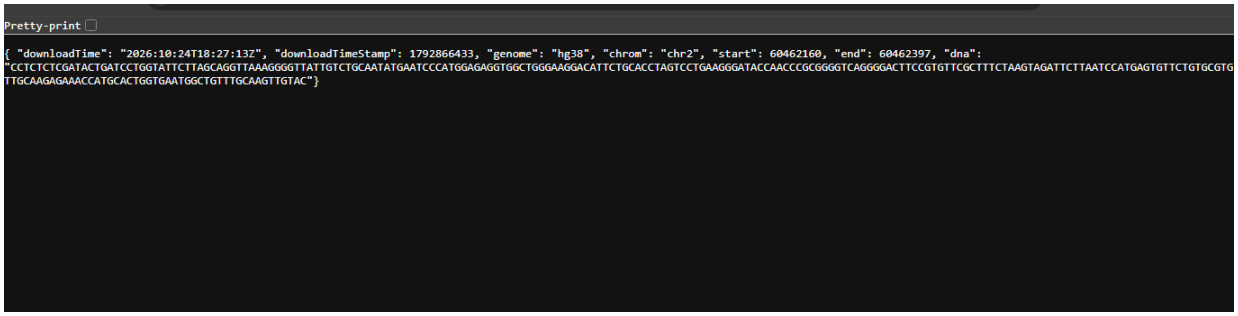


Caption. UCSC view of chr2:60,462,161-60,462,397 (237 bp, hg38). The CHOPCHOP ranked targets appear as coloured bars over the region.

What it shows. The position box shows exactly 237 bp. The coordinates come from the primer table: the right primer starts at 60462161 and the left primer ends at 60462397.

Why it matters. This is the region that one PCR product covers, so the same sequence supports the CRISPR, TALEN and ZFN designs.

Figure Z4: The sequence returned by the UCSC sequence service



Caption. JSON output for hg38 chr2, start 60462160, end 60462397, retrieved 24 Oct 2026. The 'dna' field holds the sequence.

What it shows. A 237 nt string beginning CCTCTCTCGATACTGATCCTGG and ending GTTGCAAGTTGTAC. It is already in the same orientation as the CHOPCHOP results, so it was used without reverse-complementing.

Why it matters. This is the raw data for the ZFN scan. We verified that it contains the TALEN site and the CRISPR spacer before using it.

5. The scan: method and result

Method. Every 9 bp window on the top strand was tested as a left half-site. For each window, a right half-site on the opposite strand was tested after a spacer of 5, 6 or 7 bp. A half-site was accepted only if it had the form GNN-GNN-GNN, the pattern most often used for modular assembly. This rule is quoted from general knowledge and should be confirmed in your designer-nuclease source.

Result. Three G-rich 9-mers were found on the top strand and one on the bottom strand. Only one combination satisfied the spacer rule.

Table 1. Candidate ZFN pair (manual scan, not a tool output)

	Left half-site	Spacer	Right half-site
Sequence (5' to 3' on its own strand)	GTGGCTGGG	AAGGACA (7 bp)	GGTGCAGAA
Triplets (fingers)	GTG, GCT, GGG	-	GGT, GCA, GAA
Strand	Top	-	Bottom (read on the opposite strand)
Genomic position (hg38)	chr2:60462236-60462244	60462245-60462251	chr2:60462252-60462260
Finger modules	(to be taken from the archive)	-	(to be taken from the archive)

Figure Z5: Candidate pair on the amplicon sequence

```
  1 CCTCTCTCGATACTGATCCTGGTATTCTTAGCAGGTTAAAGGGGTTATTGTCTGCAATAT
 61 GAATCCCATGGAGAGGTGGCTGGGAAGGACATTCTGCACCTAGTCCTGAAGGGATACCAA
121 CCCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGT
181 TCTGTGCGTGTTGCAAGAGAAACCATGCACTGGTGAATGGCTGTTTGCAAGTTGTAC
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Caption. The 237 nt amplicon (numbered from 1). Blue = left half-site, grey = 7 bp spacer, red = region read by the right half-site (it is read on the opposite strand, as GGTGCAGAA). **Green bold** = the TALEN pair site used earlier (positions 107-157). The CRISPR spacer starts at position 105.

Why it matters. It shows that the ZFN candidate lies upstream of, and very close to, the CRISPR and TALEN sites.

6. How the candidate relates to the other designs

Design	Position (hg38)	Notes
ZFN candidate	chr2:60462236-60462260	9 bp + 7 bp spacer + 9 bp; 3 fingers per side
CRISPR guide	starts chr2:60462265	about 5 bp from the end of the ZFN right site
TALEN pair (Rank 3)	starts chr2:60462267	18 bp + 15 bp + 18 bp; about 30 bp from the ZFN left site

7. What this shows for the paper

- **Site availability is restricted.** With the GNN rule, one pair exists in 237 bp. CRISPR and TALEN designs were available across this region.
- **Complexity.** One ZFN pair needs about 6 fingers (3 per side) plus two FokI fusions, and each finger must still be tested for binding.
- **Contextual interference.** Adjacent fingers influence each other's binding, so a finger that works alone may bind poorly next to other fingers. In-silico prediction is therefore unreliable, and success rates are lower than for CRISPR or TALENs. Back this statement with numbers from your PDF.
- **Honest framing.** Say the pair is a candidate from a manual scan, to be confirmed with ZiFiT or ZFNGenome when those tools are reachable.

8. Three-way comparison

	CRISPR/Cas9	TALEN	ZFN
Recognition unit	~20 nt guide RNA	1 base per repeat	3 bp per finger
Parts per target (this exon)	1 guide	36 repeats (2 proteins)	about 6 fingers (2 proteins)
Candidates in this region	several	several	1 under the GNN rule
Main design difficulty	low	repetitive assembly	contextual interference
Position	chr2:60462265	chr2:60462267	chr2:60462236-60462260

Rate predictability from your source PDF rather than from this table.

9. Limitations

- The pair came from a manual scan with one rule (GNN x3, 3 fingers per side, 5-7 bp spacer). Other rules and finger numbers would give more candidates.
- No tool score, off-target prediction or module list was obtained. Off-target risk for ZFNs is therefore not assessed here.
- The finger numbering depends on binding direction (a finger reads its triplet in the opposite direction to the protein). Check the convention in your source before numbering.
- Coordinates are hg38. The sequence service returned 237 nt, one base more than the 236 bp product reported by the primer table; this is a counting difference between the two systems.

10. Still to do before submission

- Retry ZiFiT (bindr.gdcb.iastate.edu/ZiFiT) from another network or browser, or use ZFNGenome, and replace the manual result if a tool pair is obtained. I could not confirm that either is online.
- Fill in the finger modules from the Zinc Finger Consortium archive or Wright et al. (2006).
- Look up and cite the references yourself: Sander et al. 2007 and 2010 (ZiFiT), Wright et al. 2006 (modular assembly), Reyon et al. 2011 (ZFNGenome).
- Check every statement about ZFN success rates and contextual interference against your designer-nuclease PDF.

- Add the access date (24 Oct 2026) to every figure caption and use the word 'predicted' for design results.

Appendix: Which screenshot is used where

Screenshot	Content	Use
403 error page	Blocked design-tool link	Figure Z1
UCSC gene view with coordinates	hg38, 103,136 bp region	Figure Z2
UCSC 237 bp view with position box	chr2:60,462,161-60,462,397	Figure Z3
Sequence service output	237 nt DNA string	Figure Z4
UCSC gene tracks without position box	Conservation and expression tracks	Not used (no position information)
UCSC 237 bp view, lower half and earlier copy	Gene and variant tracks	Not used (duplicates Figure Z3)